

Do structure-predicted solvent responses identify where alchemical simulations need more sampling?

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Abstract

Alchemical free-energy calculations distribute simulation effort across a set of intermediate Hamiltonians. Equal allocation is simple, but inefficient when sampling difficulty varies strongly between windows. We asked whether the fifteen structure-predicted solvent-response coordinates in SolvAI could identify difficult windows before or during a calculation, beyond diagnostics derived from the target trajectory itself. We preserved the fixed 15-point integration grid and preregistered a molecule-held-out, zero-new-simulation identifiability gate before considering new physics. Across twelve molecules and 180 PIMD2 windows, early variance diagnostics modestly predicted disagreement between later complementary blocks. A generic observed-data model reduced log-difficulty error from 1.547 to 1.427, but the SolvAI-conditioned model increased it to 1.475 and was indistinguishable from molecule-shuffled controls. The SolvAI increment was absent at every fixed prefix between 0.5 and 3 ps. Independent-reference planning further showed that the first predeclared design with adequate power and reference precision would require 256 molecules, two full-grid streams and 3 ns per stream-window, approximately 69,138 reserved RTX 3090 GPU-hours under optimistic noise scaling. No new simulation was therefore launched. The result separates two forms of transferable physics: response coordinates that improve an experimental hydration endpoint need not predict the sampling difficulty of the underlying alchemical dynamics.

1 Introduction

Adaptive simulation is attractive because different alchemical windows can have very different correlation times, transients and estimator variances. An allocator that recognizes those differences early could devote additional trajectory time where it most reduces uncertainty while retaining a conservative integration grid. The key question for Active SolvAI is stricter than whether adaptive sampling works: molecular response information predicted before simulation must improve allocation beyond a competent policy that already observes the target trajectory.

Earlier Active SolvAI studies addressed a different allocation variable. They tested molecule-specific selection of lambda locations and found that a SolvAI-conditioned Bayesian-quadrature policy lost to uniform direct integration. An apparently strong same-curve oracle also collapsed when lambda positions were selected and evaluated on complementary trajectory blocks. A proposed independent-replica resolution experiment was stopped at its prospective power gate. Those negative results rule out a simple return to placement optimization, but do not establish whether all fixed windows need equal trajectory length.

We therefore isolated a new estimand: all policies use the same Hamiltonian, fifteen lambda values, quadrature and mandatory initial sampling floor; only the allocation of additional immutable trajectory chunks may differ. Before spending new GPU time, we asked whether the requisite target—future window-level sampling difficulty—is identifiable from existing trajectory prefixes, and whether frozen SolvAI responses add information beyond generic observed diagnostics. This

preregistered gate produced a negative answer for the SolvAI increment and a prohibitive lower-bound design for a definitive independent test.

2 Results

2.1 Inherited evidence fixes the test

All parent SolvAI and Active SolvAI results were regenerated from machine-readable artifacts. The parent response layer improves the matched experimental hydration model from 0.303 to 0.202 kcal mol⁻¹ on the ARROW reference set, but the prior Active program found no endpoint gain from actual three-point PIMD2 responses. On eight dense sentinels, Active Bayesian quadrature produced integral MAEs of 1.701 and 1.608 kcal mol⁻¹ at five and seven windows, compared with 1.153 and 1.092 kcal mol⁻¹ for uniform direct integration. Cross-block oracle selection was worse than uniform by 0.431 and 0.409 kcal mol⁻¹. These results were retained without reinterpretation.

The trajectory census found 363 qualified PIMD2 response files, including 72 molecules with complete observations at lambda 0.1, 0.5 and 0.9. Only twelve molecules had all fifteen windows, and each molecule-window had one 5-ps stream. Contiguous portions of these trajectories can diagnose identifiability but cannot provide independent-replica validation.

2.2 Generic diagnostics carry modest signal

For each of the 180 dense molecule-windows, the preregistered primary analysis exposed the first 1 ps and predicted a stabilized measure of disagreement between two non-overlapping later 1-ps blocks. All preprocessing and ridge-penalty selection were nested within leave-one-molecule-out validation. The generic model used lambda, quadrature weight and prefix diagnostics; the matched SolvAI-conditioned model added the fifteen frozen response coordinates and fixed lambda interactions.

The lambda/protocol baseline had a log-difficulty MAE of 1.547. Generic trajectory diagnostics lowered this to 1.427, a paired difference of -0.120 (90% confidence interval, -0.199 to -0.041). The 7.7% reduction was statistically directional but missed the preregistered 10% materiality threshold. Variance and SEM-like statistics were the most informative individual diagnostics, with rank correlations of approximately 0.32–0.36 with later complementary disagreement.

2.3 SolvAI conditioning does not improve allocation information

The SolvAI-conditioned model had an MAE of 1.475, 0.0477 higher than the generic model (paired 90% confidence interval, -0.0171 to 0.1199), with six of twelve molecules improved (Fig. 1). Mean within-molecule rank correlation was 0.225, below the frozen 0.30 criterion. The aligned model differed from the mean of five molecule-shuffled controls by only -0.0102 (90% confidence interval, -0.0478 to 0.0310). SolvAI conditioning was also worse than generic diagnostics at 0.5, 2 and 3 ps. SolvAI coordinates alone performed worse than lambda/protocol by 0.129 (90% confidence interval, 0.051 to 0.225).

These controls distinguish a failure of the proposed molecule-conditioned increment from a total absence of time-series information. Generic diagnostics had modest value; the frozen response representation did not identify additional window-specific difficulty.

2.4 A resolving independent design is impractical

We next ran a separately frozen simulation-based power and reference-reliability calculation rather than launching a small campaign. The minimum-useful alternative was a 20% error reduction

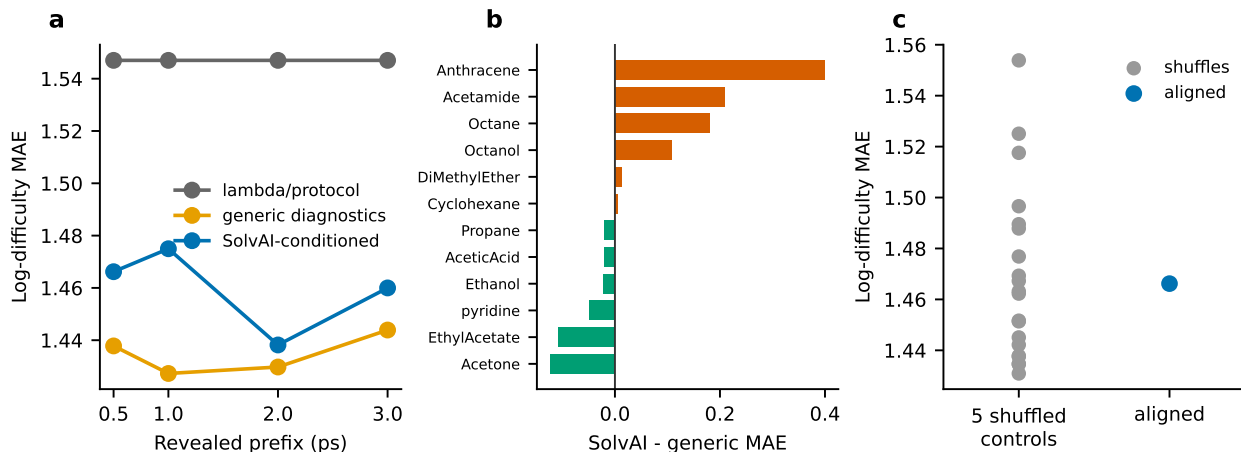


Figure 1: **Frozen SolvAI responses do not add identifiable sampling-difficulty information.** **a**, Molecule-held-out error across all fixed prefix lengths. **b**, Molecule-level change in absolute prediction error at the primary 1-ps prefix; negative values favor SolvAI conditioning. **c**, The aligned response model compared with five preregistered molecule-shuffled controls. All outcomes use complementary blocks from one short trajectory and are development diagnostics, not independent-replica validation.

over the inherited generic policy scale. Molecule-level effect heterogeneity came from the eight development-exposed v1 cases, and independent reference disagreement was propagated from the v2 complementary-block audit using optimistic inverse-square-root time scaling.

The first adequate point in the predeclared grid used 256 molecules, fifteen windows, two independent streams and 3,000 ps per stream-window (Fig. 2). This is 23.04 microseconds of production, projected as 62,287 RTX 3090 GPU-hours or 69,138 reserved hours including contingency. A 20-molecule, 50-ps campaign would have powers of only 0.326 and 0.142 at the two fixed budgets and zero probability of passing the reference criterion. No new trajectory was launched.

3 Discussion

This bounded study rejects the current Active SolvAI v3 hypothesis at its first causal gate. SolvAI response coordinates are useful for the experimentally supervised hydration endpoint, but that usefulness does not transfer automatically to sampling difficulty. The two targets depend on different information: endpoint response priors describe solvent thermodynamics, whereas effort allocation depends on correlation, relaxation and finite-time estimator behavior under a particular simulation protocol.

The generic result is informative but insufficient for a new claim. Early variance statistics modestly anticipate later disagreement, suggesting that ordinary adaptive sampling could be useful with long independent development trajectories. Yet generic adaptation alone is not Active SolvAI, and it missed the frozen materiality threshold on the inherited short data. The correct conclusion is therefore a no-go for the present molecule-conditioned formulation, not evidence that adaptive sampling is impossible.

The main limitation is measurement. One 5-ps stream cannot establish an independent sampling-difficulty target, and inverse-square-root extrapolation may be optimistic in the presence of slow modes. That limitation does not justify an underpowered pilot: the frozen planning calcu-

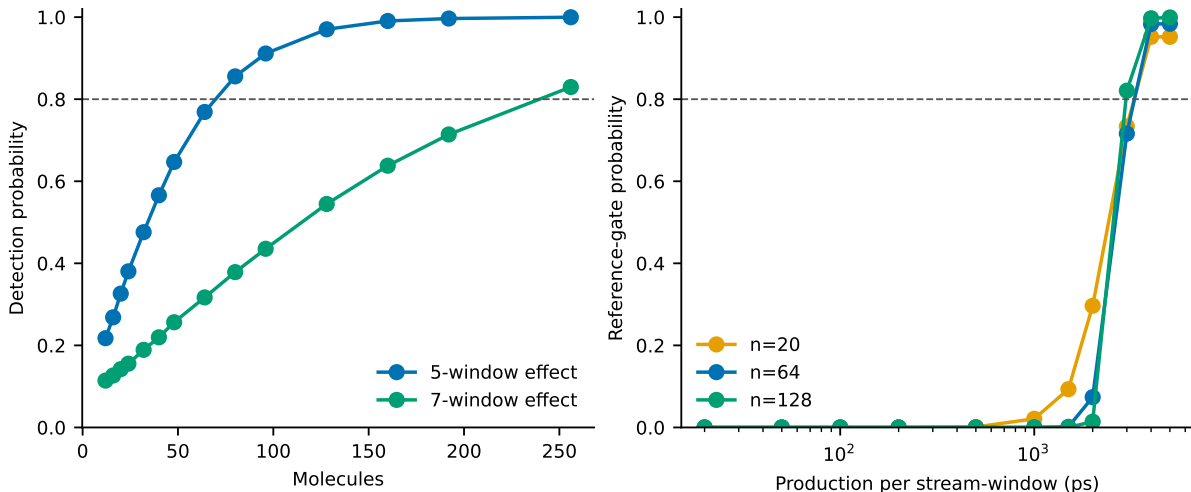


Figure 2: **Power and reference precision jointly rule out a bounded campaign.** Left, probability of detecting a 20% SolvAI-over-generic effect under inherited molecule-level heterogeneity. Right, probability that the cohort 90th percentile of dense-reference disagreement meets the frozen 0.111 kcal mol⁻¹ threshold under optimistic inverse-square-root scaling. Dashed lines mark the 0.80 launch gates.

lation makes the required scale explicit. A future program would need a response representation trained specifically on protocol-level sampling difficulty across hundreds of independent full-grid trajectories. It would be a new research program rather than a continuation or rescue of the present test.

4 Methods

4.1 Prospective governance

The v3 blueprint was hashed and read in full. The inherited evidence audit and master protocol were committed before Gate-1 scoring; a pre-result amendment clarified fixed lambda interactions needed for molecule-conditioned linear predictions. The power and reference protocol was committed after Gate 1 but before its Monte Carlo calculation. All prior molecules and results were marked development-exposed. No experimental hydration label entered a v3 predictor, policy, stopping criterion or selection rule.

4.2 Trajectory processing

We read native Arbalest ‘SYSTEM dHdL’ values in kcal mol⁻¹, removed the time-zero record and retained 50 observations at 0.1-ps intervals per window. Lambda values were ordered numerically and integrated with the fixed trapezoidal weights. Each stream was partitioned into ten 0.5-ps blocks. The primary prefix used frames through 1.0 ps; the target was

$$\log \left[\frac{1}{2} (\bar{y}_{3.1:4.0} - \bar{y}_{4.1:5.0})^2 + 0.25^2 \right].$$

Sensitivity analyses used 0.5-, 2- and 3-ps prefixes and stabilizers of 0.10 and 0.50 kcal mol⁻¹.

4.3 Diagnostics and response features

Prefix summaries included the response mean and variance, naive SEM, non-overlapping and overlapping batch variance rates, a Bartlett–Newey–West variance rate, lag-1 autocorrelation, initial-positive-sequence autocorrelation time, effective sample size, half-prefix location and scale differences, Kolmogorov–Smirnov distance, linear/rank drift, temperature, density and unresolved-equilibration flags. SolvAI coordinates were generated by the immutable released structure-to-response surrogates. No source calculation was rerun.

4.4 Nested prediction and controls

Ridge penalties 0.1, 1, 10, 100 and 1,000 were selected inside each outer training set using leave-one-training-molecule-out error. Imputation and standardization were fit on training rows only. Molecule-level coordinates entered as main effects and fixed interactions with centered lambda and its square. Five joint molecule-level response permutations used seeds 88031–88035. Primary uncertainty used 100,000 molecule- clustered bootstrap resamples with seed 20260903.

4.5 Power and cost

The power calculation ran 50,000 Monte Carlo trials per condition. It centered the eight inherited paired policy effects and added null, 20% and 30% alternatives. Detection required a two-sided 95% paired interval excluding zero. Reference disagreements were resampled from the 24 unique 2.5-ps complementary partitions and scaled by the square root of inverse duration. Production cost used the measured 369.902 PIMD2 ps GPU⁻¹ h⁻¹ throughput; the operational reservation added 11% for failures and restarts.

Data and code availability

The branch ‘active-solvai-v3-effort-allocation’ contains every freeze, raw-file hash, molecule-level prediction, decision ledger, analysis script and compiled report. Earlier SolvAI and Active SolvAI histories remain immutable. No new simulation data were generated in v3.